House Price Prediction Model

Objective:

- ➢ Understand the Dataset & cleanup (if required).
- ➢ Build Regression models to predict the sales w.r.t a single & multiple feature.
- ➢ Also evaluate the models & compare their respective scores like R2, RMSE, etc

Introduction

The house price prediction project aims to develop and evaluate a machine learning model capable of predicting house prices based on various features. This project is particularly challenging due to the presence of strong multicollinearity among features. By addressing this challenge, the project seeks to build a robust predictive model.

Dataset:

- ➢ **Source:** https://www.kaggle.com/datasets/yasserh/housing-prices-dataset?select=Housing.csv
- ➢ **Size:** 10,000 records
- ➢ **Features:** Includes information about bedrooms, bathrooms, square footage, location, condition, and more.
- ➢ **Target Variable:** price

The dataset provides a comprehensive overview of housing characteristics and their corresponding prices, making it suitable for regression analysis.

Data Preprocessing

Data preprocessing is a crucial step to ensure the quality and reliability of the model. It involves the following tasks:

- **Date Formatting**: Converted the date column to a datetime format and set id as the index to facilitate time-based analysis and ensure record uniqueness.

- **Data Type Conversion**: Converted columns such as price, bathrooms, and floors to integer types to align with their discrete nature.
- **Feature Engineering**:
    - **House Age**: Calculated house_age by subtracting yr_built from the year extracted from the date column to quantify the age of each house.
    - **Renovation Indicator**: Created a binary feature renovated indicating whether a house was renovated based on yr_renovated.
- **Dropping Columns**: Removed date, yr_renovated, and yr_built as they were either redundant or transformed into other features.
- **Handling Missing Values**: Conducted a thorough check for missing values and found none, ensuring data completeness.
- **Duplicate Check**: Verified the uniqueness of records, confirming no duplicates were present, thus maintaining data integrity.

Exploratory Data Analysis

EDA was performed to gain insights into the dataset and understand feature distributions:

- **Descriptive Statistics**: Calculated summary statistics, including mean, median, and standard deviation, for all numerical features to understand their central tendencies and variability.
- **Distribution Plots**: Visualized the distribution of numerical features using histograms and calculated skewness to identify any non-normal distributions that might require transformation.
- **Count Plot**: Created a count plot for the grade feature to assess its distribution and identify any dominant categories.
- **Box Plot**: Used a box plot to explore the relationship between grade and price, highlighting outliers and potential trends.
- **Correlation Matrix**: Calculated and visualized correlations between features using a heatmap to identify multicollinearity and determine the strength of relationships between features.

Feature Selection

Feature selection was conducted to identify the most relevant features for predicting house prices:

- **Recursive Feature Elimination (RFE)**: Employed RFE with a Linear Regression estimator to select the top 5 features based on their importance scores, which aids in reducing overfitting and improving model performance.

Model Development

Multiple regression models were developed and trained using the selected features:

- **Linear Regression**: A simple yet powerful model to establish a baseline performance.
- **Random Forest Regressor**: Utilized for its ability to handle non-linear relationships and interaction effects between features.
- **Decision Tree Regressor**: Provides insights into the decision-making process of the model and serves as a baseline for ensemble methods.
- **Ridge Regression**: Implemented with L2 regularization to handle multicollinearity and reduce model complexity.
- **Lasso Regression**: Employed with L1 regularization to enforce feature sparsity and automatically select features.

Model Evaluation

➢ **Metrics:** Evaluated models using Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared.

➢ **Comparison:** Compared the performance of different models based on the evaluation metrics.

Best Fit Model

Random Forest Regressor: Achieved the best performance with an R-squared of **85.41%**.

Model Visualization

- **Scatter Plot:** Plotted actual vs predicted prices for the Random Forest model.
- **Residual Plot:** Visualized the residuals of the Random Forest model.

Conclusion

The Random Forest Regressor model demonstrated the highest accuracy in predicting house prices. This model can be utilized for estimating house prices based on the selected features.

Future Improvements

- **Hyperparameter Tuning:** Fine-tune the hyperparameters of the Random Forest model to potentially improve performance.
- **Feature Engineering:** Explore additional feature engineering techniques to extract more informative features.
- **Ensemble Methods:** Consider combining multiple models using ensemble methods to further enhance prediction accuracy.